take the first photograph tore the very electron of the atom itself out of its intra-atomic bond, so that the atom appearing on the second picture would never, in any circumstances, possibly be the same, unchanged atom. At best, we would discover the electron somewhere far outside, far away from the nucleus.

It is therefore, evidently, fundamentally impossible to observe the orbit of an electron within the atom. But this impossibility is not due to any shortcoming (still remediable) of that postulated ideal microscope, assumed to be as perfect as natural laws would allow it to be, but rather a consequence of those very laws. By the operation of these self-same natural laws, extremely violent methods—as exemplified by the use of an electron microscope, where the electrons are accelerated in their orbits by a very high voltage—are our only means of getting a sharp, well focused picture of an atom which is in conformity with natural laws.

At this point, it can no longer astonish us to discover that we have reached the limits of visualization, and that the concept of electrons circling a nucleus cannot be taken literally, like the concept of a water molecule consisting of two hydrogen atoms and one ogygen atom, arranged in a triangular pattern.

The limitations imposed here on visual representation can be formulated with greater accuracy with the aid of a relationship, called the *uncertainty principle,* which is based on the quantum theory. It can be expressed in the simplest form as follows: One can never know simultaneously with perfect accuracy both of those two important factors which determine the movement of one of these smallest of particles—its position and its velocity. It is impossible to determine accurately *both* the position and the direction and speed of a particle *at the same instant.* If we determine experimentally its exact position at any given moment, its movement is disturbed to such a degree by that very experiment that we shall then be unable to find it at all. And conversely, if we are able to measure exactly the velocity of a particle, the picture of its position becomes totally blurred.

But by this statement I have anticipated the climax of a development which I must now attempt to describe from its very beginning.

Quite apart from the difficulties involved in obtaining a visual representation, the atom model has certain other properties which do not seem to agree at all with actual experience. For, according to all our previous knowledge, an electron revolving around an atomic nucleus should not be able to revolve in a circular or elliptical orbit for any length of time. In the first place, the electron carries an electric charge, and secondly, it vibrates in its orbit around the nucleus. As the vibrations of electrons in a radio aerial produce an electric wave, the electron vibrating in the atom must emit a wave of radiation, which in this case we should observe as ultraviolet light. But this would mean an emission of energy at the expense of the planetary electron,